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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	#slaynergy
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, 5, 8, 9, 10
Project / Solution Name	Jobbery
Name (Leader)	Yang Zi Chien
Email & Phone (Leader)	zichienyang@gmail.com
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Outside of personal preference towards Artificial Intelligence (AI), it is undeniable that AI heavily impacts human lives. It enables automation, significantly reducing the demand for labour in various industries, including logistics and customer service. Despite the numerous benefits of AI, one major concern is its impact on human jobs.

According to Mohd Arshad (2024), 1.6 million workers were projected to be replaced by AI by 2026, particularly in high- and middle- skill occupations. In contrast, low-skill occupations show a lower displacement impact. On paper, Malaysia's graduate unemployment rate looks healthy at just 3.2% (DOSM, 2024). In reality, many graduates are taking non-technical jobs. As AI displaces more skilled roles, finding a job that actually uses their degree will only become tougher.

Furthermore, Malaysia ranked 83rd out of 130 countries in digital skills (World Economic Forum, 2023), indicating an underestimated skill gap and lack of awareness of AI adaptation, as supported by the data that 68% of Malaysian firms lack training and development in AI adaptation (MDEC, 2022), leaving HR and L&D managers without clear evidence of which skills to build in their workers.

This shows that reskilling and upskilling involve all working adults, graduates, employers and educators. Therefore, the key question is: how might we give Malaysian workers, students, employers and educators a clear view of how AI will reshape jobs, and guide them towards the right skills and future roles, so that both individuals and the nation can close the AI skills gap and thrive in an AI-driven economy?

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Jobbery is a web application that helps Malaysian workers see how AI will reshape their careers, and act before their skills lose relevance. Users upload a résumé or answer a few short questions, then Jobbery turns their profile into a matched future roles and a reskilling plan, with every role and skill linked to the UN Sustainable Development Goals (SDGs).

Jobbery forecast which skills will rise, decline or remain. From open data sources on AI capability (Epoch AI benchmarks), AI exposure by occupation (ILO, Anthropic Economic Index) and Malaysian job advertisement trends (DOSM). From these, it estimates when AI could perform each skill under slow, medium and fast AI-progress scenarios. These AI progression forecasts (slow, medium and fast AI progression) are strengthened by a prediction market, where users use free tokens to forecast the future skill trend. Accurate forecasters earn a bigger vote, and the crowd's judgement updates how likely each scenario is, adding human insight to the model's predictions.

Jobbery then matches each user's skills and preferences against job, pay and unemployment data. This shows where opportunities cluster across Malaysia, such as job opportunities in semiconductor companies in Penang. Finally, it combines the user's résumé, skill forecasts and regional information to rank reachable roles. It then builds a personalised reskilling pathway from verified free and low-cost courses, including HRD Corp and PERKESO programmes.

Through the SDC Phase 1 workshops, we learned how to use Replit Core AI for agent-assisted application development, turning our project idea into a working prototype. We applied these skills to generate the initial web application structure, develop the interface and refine features through iterative prompting. This gave us a foundation for Jobbery's résumé analysis, career matching and personalised learning plans. Our team reviewed, tested and revised the generated code to ensure it met our requirements.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies we use to build the deployed prototype:

Web app: Next.js 16 and React 19 (written in TypeScript), styled with Tailwind CSS. Charts use Recharts.

Forecasting engine: it reads the dated data files and forecasts each skill's fading time. It keeps three possible speeds of AI change (slow, medium and fast) and weights them with a statistical method called Bayesian model averaging. Each speed is scored on how well it predicted past official releases, older results count for less.

Prediction market: a points-only market (never real money) asks checkable questions about the next DOSM and Epoch AI releases, and its prices are set by a standard market-maker rule (LMSR). It is seeded with 200 simulated forecasters, always labelled as simulated. When new data arrives, the questions settle.

Jobs Matching and Planning engine: it suggest only the jobs the user can reach, job demand, time to reach it, and fit with their interests. It then builds the week-by-week plan from the skills still missing, using hand-checked free or low-cost courses tailored to the user's weekly hours.

GenAI API: the Groq API running Qwen3.8 27B, an open-weight model, used only to read resumes.

Database: PostgreSQL, hosted on Neon, holds each visitor's sessions and progress, advisor bookings, and a tamper-evident activity log.

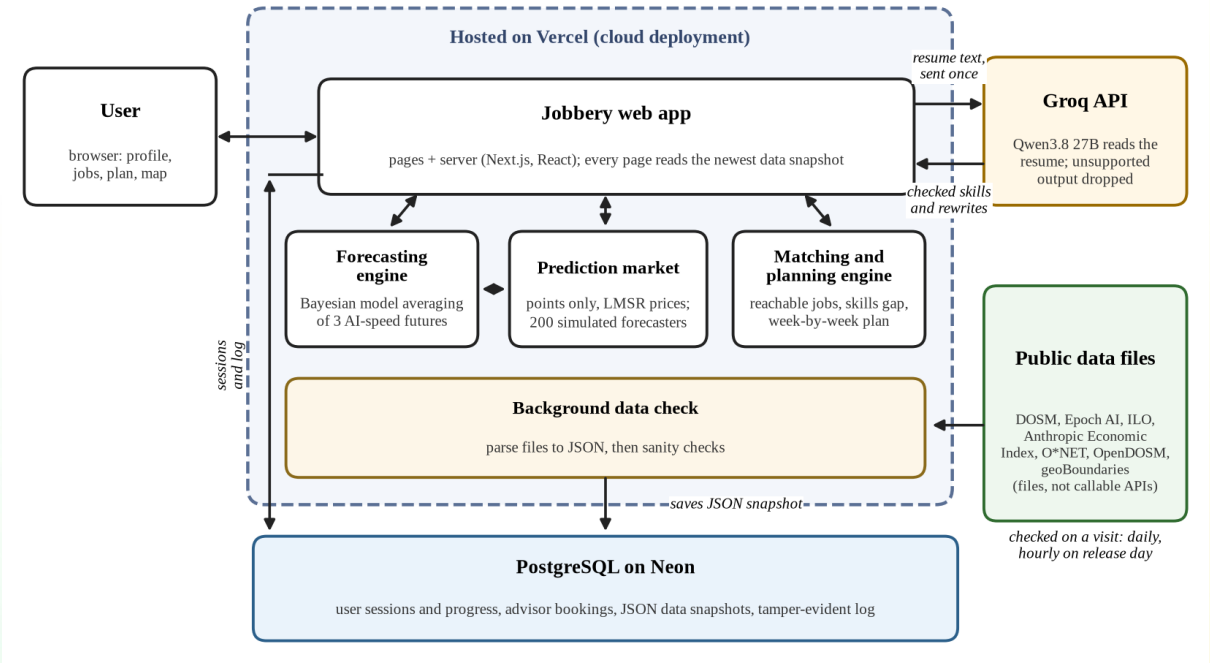
Data: dated public files (DOSM, Epoch AI, ILO, Anthropic Economic Index, O*NET, OpenDOSM, geoBoundaries), downloaded and parse to JSON format.

Hosting: Vercel deployment. The code is on public GitHub under the Apache-2.0 licence.

How the pieces connect: the users interacts with Jobbery web app on Vercel. The app asks the engine for forecasts, matches and plans, saves progress in PostgreSQL, and, only if the user consented to it, sends the resume text once to Groq to be read. Diagram below.

Figure 1. System block diagram.

Did the project use an official SDC sponsor-provided technologies? Yes, we use replit core AI to help with the code generation. Runpod however is not used because nothing in Jobbery needs a GPU.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Through brainstorming and research findings, our team proposed an AI-powered career development platform that utilises existing AI technologies to analyse uploaded CVs, identify skill gaps, and recommend relevant online courses. We also incorporated a prediction market where users can vote on future AI-related trends, submit their own predictions, and receive recognition for accurate forecasts once the outcomes are confirmed. Professional career

counseling through appointment bookings was also proposed to provide personalised guidance.

During prototyping, we developed a functional website with AI-powered CV analysis and course recommendations, which demonstrates the technical viability of these core features. To improve user-friendliness, we designed a clear homepage explaining the available services, provided step-by-step instructions and illustrations, and included real-life graphs to present career-related trends and statistics. However, the prediction market and AI-powered CV analysis require further validation to assess AI recommendation accuracy, prediction reliability, and the practical implementation of counseling sessions. Therefore, Phase 2 will focus on refining these features, gathering user feedback, and addressing limitations in real-world data and available resources to improve the platform’s overall reliability and effectiveness.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	1. Target 9.2 "Promote inclusive and sustainable industrialisation" 2. Target 10.3 "Ensure equal opportunities and end discrimination" 3. Target 4.4 "Increase the number of people with relevant skills for financial success" 4. Target 5.7 "Equal rights to economic resources, property ownership, and financial services" 5. Target 8.5 "Full employment and decent work with equal pay"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	"The findings show that 697,000 jobs are expected to be highly affected by AI, digitalisation and the green economy if workers do not upskill within the next three to five years." – Human Resources Minister Datuk Seri R. Ramanan, 24 June 2026
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Percentage (%) increase in website users. 2. Percentage (%) of users still relying on the provided AI resources. 3. Percentage (%) of users who have completed the AI courses and passed the AI quizzes. 4. User testimonials on whether or not they have found jobs within 1-3 years of using the website, and what they think

Metric Category	Your Response
	are the specific contributing factors (AI courses, personalised job matching, etc.)
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Setting a Phase 2 test period of 5 years, Jobbery is predicted to obtain these quantitative results: 1. 35,000 website users. 2. 90% of users still relying on the provided AI resources. 3. 50% of users have completed the AI courses and passed the AI quizzes. 4. 40% of users who have utilised the AI resources can find employment within 3 years, sourced from the website's feedback section.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

Yes ☑️ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑️ Yes No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16.4.2; React 19.3.0	Code	MIT
Tailwind CSS 4.3.3; Framer Motion 14.4.0; Recharts 3.10.1	Code	MIT
Lucide React 1.60.0	Code	ISC
TypeScript 5.9.3	Code	Apache-2.0
Node.js 22	Code	MIT
pg 8.23.1; better-sqlite3 13.0.3; fflate 0.8.3	Code	MIT
unpdf 1.9.1 (reads PDF resumes)	Code	MIT
mammoth 1.13.0 (reads Word resumes)	Code	BSD-2-Clause
Instrument Sans (font, via Google Fonts)	Code (font)	SIL Open Font Licence 1.1

Qwen3.8 27B (resume reader, via the Groq API)	Model	Apache 2.0 (open weights)
DOSM Job Market Insights, 2026 Q2	Dataset	Malaysian Government Open Data Terms of Use 1.0
OpenDOSM employment, unemployment and wage data by state	Dataset	CC BY 4.0
Epoch AI benchmarking data	Dataset	CC BY 4.0
Anthropic Economic Index (2025-02-10 release)	Dataset	CC BY (data)
ILO generative-AI exposure scores (2025 update)	Dataset	CC BY 4.0
O*NET occupation and task data	Dataset	CC BY 4.0
geoBoundaries Malaysia state borders (OpenStreetMap)	Dataset	ODbL 1.0
UN SDG target text (sdgs.un.org)	Dataset (text)	Quoted with source

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

Yes ☑️ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Name: Yang Zi Chien
24 Oct 2026